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1. Purpose & Scope

1.1. General

The intent of this document is to specify the mandatory requirements and engineering best practices in relation to machine in mechanical design that meet Nestlé's basic TPM (Total Performance Management) requirements for machine and hygienic design in order to ensure product quality and safety.

This specific module covers the **Basic, Medium and High level Requirements**, for **hygienic design**.

1.2 Compliance

Machine must comply with the following:

- All relevant European or United States of America machine directives and standards;
- Local legal requirements, and local machine standards (i.e. in the country of installation)

This standard is not intended to replace any local machine regulations or requirements. Local requirements must be met in addition to the requirements of this standard. In the event of conflicting requirements, the most stringent requirements must be met.

The supplier is responsible for identifying and utilizing all relevant directives or codes, standards and legal requirements. The supplier must provide a document with a list of standards used and certifying conformance by the supplier.

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Purpose & Scope

1.3 Inclusions

This document is one part of a complete specification package and may not be used unless all other documents are included.

The specification package consist of the following modules:

- Module 1 - Project & Line Specification - URS
- Module 2 - Machine Specification - URS
- Module 3 – Commercial
  - Terms and Conditions
- Module 4 - Safety, Health & Environment (SHE)
  - Machinery Safety Checklist
- Module 5 – Mechanical and Hygienic Design
- Module 6 – Electrical & Automation (E&A)
- Module 7 – Acceptance
  - Factory Acceptance Test (FAT) Checklist
  - Site Acceptance Test (SAT) Checklist

Additional local requirements may be specified as an addendum to each module or as a single module with corresponding chapters to the modules listed above.

- Module 8 – Local Requirements (if required)

1.4 Deviations

Any deviation from the requirements of this standard must be submitted by the supplier in writing to Nestlé. Receipt of this deviation list does not constitute acceptance of exceptions to the standard. Deviation requires written approval from the Nestlé project manager.

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2. Terminology & Definitions

2.1. Basic Hygiene Requirements	Area where products are already protected in their final packaging; such areas can be at the end of a packaging line, or the finished goods warehouse.
2.2. Medium Hygiene Requirements	a. For primary filling & packing and non-sensitive products. For example: • primary packing of noodles, chocolates, culinary, coffee, breakfast cereals, pet care, … • cartoning machines with spoon inserter (as spoon will be, after opening, in direct food contact), … b. Those machines are normally installed in "Medium Hygiene areas" as defined in the Nestlé factory zoning plan.
2.3. High Hygiene Requirements	a. For primary filling & packing of sensitive products and/or machine that need to go through a wet cleaning process. For example: • primary packing of infant formulas and infant cereals • milk powder • water • RTD beverages • chilled dairy, … b. Those machines are installed in "High Hygiene areas" or "Wet cleaning area" as defined in the Nestlé factory zoning plan.
2.4. Aseptic	a. For primary filling of sterilized products, in a sterile pack and under sterile filling conditions. b. An aseptic machine is designed to be easy cleanable, sterilisable and impermeable to micro-organisms to maintain its aseptic status.
2.5. Product	It is any finished or semi-finished food item manufactured by Nestlé.
2.6. Auxiliary Materials	Auxiliary Materials are items that are part of the product bill of materials (e.g. measuring spoon, ice-cream sticks, straw). Auxiliary Materials are in direct contact with the product prior to or at the time of consumption. The hygienic design requirements for the equipment that handles auxiliary materials is the same as the hygienic design requirements as the machine that handles product
2.7. Product Safety	Assurance that the product will not cause harm to the consumer when it is prepared and/or eaten according to its intended use.

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2. Terminology & Definitions

2.8. Surface in Contact with Product	a. All surfaces of the machine that intentionally or unintentionally (e.g. due to splashing) come in contact with the product. b. In addition, all surfaces from which product or condensate can drain, drop or drawn back into the main product stream or the packaging material and have the potential to indirectly cross- contaminate the product. c. All surfaces immediately adjacent to product contact zones are to be designed as if they were product contact zones. A risk analysis will help define areas of cross contamination.
2.9. Hollow body	Void spaces, inaccessible to cleaning, which may become sources of contamination.
2.10. Conditions of intended use	All normal or reasonably anticipated operating conditions, including those of cleaning. These conditions should set limits for variables such as time, temperature and concentration.
2.11. Contaminant	Any biological, chemical, foreign matter or other substances not intentionally added to the product recipe, which may compromise food safety or suitability.
2.12. Contamination	The introduction or occurrence of a contaminant in the product or product environment (e.g. Filler, freezer, etc.).
2.13. Easy Access	A location that can be safely reached by a personnel from the floor, a platform or other permanent work area. A location that is reachable, visible, and safely accessible by personnel.
2.14. Cleanability	The suitability of the machine to be freed from soil easily.
2.15. Cleaning	The removal of soil, food residues, dirt, grease or other objectionable matter.
2.16. Dry Cleaning	Cleaning which does not involve any use of liquid. Dry cleaning can be conducted either manually or semi-automatically.

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2. Terminology & Definitions **Error! Use the Home tab to apply Titre 4;Map Title to the text that you want to appear here.**

2.17. Wet Cleaning	a. Cleaning which does involve the use of water with or without a mixture of chemicals. b. This procedure is carried out only when product is not exposed and using methods that limit the amount of water applied and its spread.
2.18. CIP	Cleaning-in-Place. It is an automated wet cleaning system of a line and/or individual machine in a closed circuit without dismantling.
2.19. SIP	Sterilization-in-Place. It is an automated process to sterilize the filler without dismantling.
2.20. External Foaming	It is the action of cleaning machine in place without dismantling by foaming with cleaning agents, through spraying nozzles (automatic) or foaming guns (manual).
2.21. COP	Cleaning-Out-of-Place It is the manual cleaning of dismantled machine.

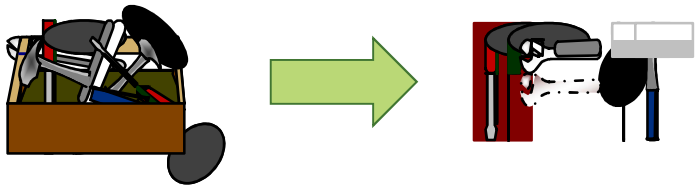
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3. 5 S

3.1. Design to support 5S

The purpose of 5S is to provide a standardised and structured process to organise and maintain the filling and packaging area. Machine must be designed to support this process. 5S stands for:

- **Sort**: Have just the necessary & remove what is not needed.
- **Set in order**: A place for everything & everything in its place.
- **Shine**: Clean by inspection & analysis.
- **Standardise**: Standardised good working practices.
- **Sustain**: Make a habit of following the 5S.



3.2. Organized tool storage

The design includes devices and accessories that allow change parts, cleaning and maintenance tools to be kept in an organized and visual way. For every necessary tool, the type of tool is specified exactly, and there is a specified storage place for the tool delivered with the machine, including cleaning and lubrication tools.

3.3. Minimum number of spare parts

The design should have a minimum number of spare parts. Regarding weights for spare parts, see Module 4 – SHE.

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4. Poka Yoke

4.1. Design for Poka-Yoke

All parts which can be removed from the machine either for cleaning, change over or maintenance should be easy to reassemble. In order not to have the possibility to assemble them in the wrong way Poka-Yoke should be applied.



Design of change parts to ensure that they can be assembled only in the correct position

5. Total Performance Management (TPM)

5.1. TPM - the Nestlé way

TPM is the basis for the Nestlé Operation Management System. This means that the machine Nestlé purchases needs to be aligned with these principles.

5.2. Supporting the TPM journey

The design of new machine for Nestlé's factories should support the Nestlé journey towards TPM:

- Autonomous Maintenance (**AM)**
- Planned Maintenance (**PM)**
- Education & Training **(ET)**

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6. DESIGN – General Hygienic Rules

6.1. Basic Principle

a. It should be noted that the machine will be installed in a food manufacturing environment and must be designed appropriately.

b. The fundamental principle involved with sanitary design is to eliminate or minimize all hollow cavities, cracks, crevices, ledges and other places which are difficult or impossible to clean.

c. Care in the design of the machine should ensure that:

- Meet the specific hygiene requirements for the type of product
- Identify and eliminate any source of contamination, being direct or indirect. If not possible, a mean to control them should be developed.
- Dust or spilled product cannot be trapped in any part of the machine.
- All surfaces should be designed to eliminate product collection or liquid pooling and must be self-draining.
- All surface areas are made of sufficient strength to prevent warping and any subsequent pooling of liquids.
- Moisture or liquids do not drip, drain, or be drawn into product zone areas.
- All parts including nonmetallic parts, will remain intact without changes in shape, structure & function through cleaning & sanitation protocols.
- Areas that accumulate or generate waste need to be identified at the design stage. Waste pans or bins due to poor machine design are not acceptable. In the event that product recovery bins or pans are required, they should remain attached and be easily accessible for cleaning or are easily removable for clean-up so that they are not lost or separated from the machine. Measures must be taken to ensure that no safety issues are created by cleaning or removing the pans.
- Belt scrapers are designed so that they do not cause or be a cause of a source of contamination. Material being scraped is collected in a sanitary manner.
- Cooling water, drain lines, drip pans (items with constant flowing water) must be piped and/or directed to a drain. These sources cannot connect directly to a drain. Path of drains cannot go directly over a product zone or its adjacent area.
- Allow for easy cleaning and inspection of all machine surfaces

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6. DESIGN – General Hygienic Rules

6.2. Basic Principle

All machinery will be installed in a food manufacturing environment and must be designed appropriately. Apply all relevant European or United States of America machine directives or codes and standards on Hygienic Design. Machinery must be compliant with USDA and FDA or European regulations for operation in a food manufacturing facility. EHEDG guidelines can also be used as a reference

6.3. Direct Food Contact

a. All material coming into contact with product, (other than stainless steel), must comply with the Nestlé standards on materials in contact with food (GI-80.008-2, the relevant abstract is included as an Annex to the present document). If no local regulations exist, beside the Nestlé standards, the material in contact with the product must comply with European (CE N°1935 / 2004 - European Directive 2011/10) or US Food, Drug and Cosmetic regulation of the United States of America (FDA CFR 21) legislations
b. For all the parts not in stainless steel, a food grade certificate is mandatory in line with Nestlé template “Certificate of Compliance for Food Contact Materials”, included as an Annex to the present document.
c. All parts in direct contact with product to be:
    - Compatible with the product characteristics.
    - Compatible with the cleaning procedure.
    - Smooth and with a non-porous finish for easy cleaning.
c. None of the materials used must release substances into the product or allow substances from the product to migrate into the material.

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6. DESIGN – General Hygienic Rules

6.4. Hollow Body

a. No hollow body shall be used for machine parts having direct food contact. This is non-negotiable.

b. On other machine parts:
   - Avoid using hollow body as the integrity of the hollow section can never be guaranteed throughout the life of the machine. Once the integrity of such hollow cavities is breached, they provide an enclosed area that cannot be cleaned and thus lead to product contamination.
   - Where the supplier believes there are no alternatives to hollow section, either due to economics or technical reasons, a clear and detailed written request, including the way parts will be assembled, should be submitted to Nestlé for approval. This must happen at the time of receiving supplier's quotation.
   - Only with Nestlé's approval, Frames can be sealed tube or sanitary channel type construction unless specific construction is requested. Frame members shall be continuously welded. Bolting shall be kept to a minimum and shall only be used for fastening solid structural members. Bolting to tubular components can only be used if weld studs are used for the tube. All penetrations of tubular components require that sleeves shall first be inserted into drilled holes through tubes and then seal welded.

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6. DESIGN – General Hygienic Rules

6.5. Niche

a. Machine is designed to prevent the ingress and accumulation of microorganisms, insects, or food residues in void or niche areas. There are no dead ends (defined as greater than 3 pipe diameters long) in any and all piping including water, compressed air, or any other utilities.
b. No piano hinges, knurling, braided covers can be used.
c. No open seams, recesses, inside threads, rivets, etc. are allowed.
d. No dead ends or spaces are permitted. All machine areas are accessible for cleaning & treatment to enable complete removal of food residues, minimization of microbiological activity and infestation by insects.
e. There are no lap joints. Examples include standing off flanged bearings versus mounting directly to side of a conveyor. Bearings should be mounted on the outside of framework and be properly spaced – they should be mounted as far from product and product contact surfaces as possible.

6.6. Glass

a. No glass should be used in any part or component of the machine. This also includes measuring devices (e.g. pressure gauge) and the HMI.
b. In some cases glass cannot be avoided and the machine supplier must receive clearance from Nestlé before freezing machine design.
c. In some other cases, whereas glass is requested by Nestlé for some specific applications, it has to be laminated glass (type of safety glass that holds together when shattered).

6.7. Wood

No wood is allowed in any part or component of the machine.

6.8. Plastics / Rubbers

a. If a material has been cleared for use in one application in the food industry, it does not automatically give blanket clearance for its general utilization. Consequently, all plastic materials should be cleared by Nestlé for each particular application.

Additional Requirements for Medium and High levels

b. In dry products, non-metallic surfaces can create electrostatic charges on the material and cause surface adhesion by small particles. This requires special attention in terms of accessibility and cleanability.

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6. DESIGN – General Hygienic Rules

6.9. Surfaces

Where technically possible, the machine design should:

- Avoid any horizontal flat surfaces, as it will accumulate dust. All surfaces to be at an angle.
- Have only smooth surfaces
- Have no crevices or ledges.

6.10. Edges & Corners

In general, sharp corners and edges must be avoided. Corners to have ≥ 6mm (1/4") radius.
All materials and components of assembly shall be free of burrs, clean and free from oil, dirt and other foreign materials.

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6. DESIGN – General Hygienic Rules

6.11. Welding

a. All welding should be:
- Lead free
- Smooth and free of voids, pinholes, cracks, splatter, etc.
- For machine located in hygienic areas, grind and polish all surfaces and welds to applicable finishes, see section Steel Quality & Finish
- All welds must be passivated.
- Scorch marks must be removed from unpainted metal surfaces
- All welding slag and splatter must be removed from any weld
- For food contact surfaces, continuous welding is mandatory
- For non-food contact surfaces, continuous welding is preferred
- Stainless steel welding to be of hygienic design (e.g. using TIG (Tungsten Inert Gas) welding).All internal corners or angles in food contact areas shall have a continuous weld and smooth radius of 6mm or greater, except that lesser radius may be used where necessary for proper functioning of parts, providing such areas can be readily cleaned. Break all edges and sharp external corners 0.08"

b. Supplier must inform which standard and /or technology they have used for welding.

Additional Requirements for High hygiene levels

c. Suppliers have to comply with the recommendations of the European Hygienic Equipment Group (EHEDG) document nº9, Sept ember 1993 “Welding stainless steel to meet hygienic requirements”.

d. Automated orbital welding is required. Exceptions are only allowed due to space limitations. All welding, independent if automated orbital or manual, must be performed under external and internal inert gas coverage.

e. Pipes must not be cut manually. The use of an electric orbital pipe saw which blocks the pipe horizontally and cut it by a rotating blade is required. If cutting by orbital saw is not possible due to space issues, a manual reciprocating saw with attached clamps for blade guidance is required.

f. Only welders of proven competence, e.g., coded for pressure vessel work (BS 5500, ASME VIII or to appropriate National Pipework Standards), are admitted to carry out welding operations on product piping.

g. The use of butt weld pipe systems for construction welding is mandatory. Prefabricated fittings are required.

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6. DESIGN – General Hygienic Rules

6.12. Stainless steel passivation for product contact surfaces	Passivation is an important surface treatment that helps to ensure the successful corrosion resistant performance of stainless steel used for product contact surfaces. a. The passivation process of equipment has to be done by the Supplier at Nestlé factory site right at the end of installation. b. The process has to be done according to procedures suggested and certified by the Supplier and reported on the technical manuals. c. To define the passivation process, the Supplier has to comply with the recommendations of the European Hygienic Equipment Group (EHEDG) document n°18, August 1998 “Passivation of stainles s steel”.
6.13. Paints & Coatings	a. Paints & coatings will be non-toxic, heavy-metal free and approved by Nestlé. b. Threaded or flanged portions, splines, and/or machined surfaces which are, or will be, in mechanical contact shall not be painted. c. Prior to painting, welds, sharp edges, burrs or other irregularities shall be ground smooth and all surfaces cleaned or paint manufacturer-approved equivalent method. d. There shall be no painted surfaces above the product stream.
6.14. Name Plates, Tags	a. Name plates and tags are minimized.
Additional Requirements for Medium	b. When attached, plates and tags are: • Continuously welded to the machinery surface • Or sealed with a joining compound.
High Hygiene levels	c. Continuously welded to the machinery surface
6.15. Legs	a. Minimize the number of legs to aid cleaning. b. Round legs are preferred over square legs. c. Threaded leg adjusters should not protrude outside of the machine and do not penetrate the frame members, see picture under 18.7.
6.16. Installation	a. Machine should not be directly bolted to the floor. b. If fixing is required, the appropriate solution needs to be approved by Nestlé.

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6. DESIGN – General Hygienic Rules

6.17. Floor Clearance	To ensure adequate cleaning and inspection, all machine shall have a ground clearance of at least 250 mm (9.84") to the lowest point of the machine frame, components and utility lines. This must not compromise Nestlé safety standards (Module 4) e.g. clearance safety fence are 180mm (7.1").
6.18. Wall Clearance	a. If machine or part of the machine (i.e. electrical cabinet) is installed on a wall, a minimum distance of 50 mm (1.9") is required between the machine and a wall to allow easy access for cleaning and inspection. b. The clearance distance will depend on size of the part to be fixed on the wall.
6.19. Overhead	Overhead installation of equipment should be avoided, as access, maintenance and cleaning are difficult.
6.20. Condensation	a. Provisions must be made to eliminate condensation on the machine. If unavoidable, the condensate must be diverted away from the product. b. Machine supplier must design the machine according to room environment conditions that are defined in the Machine Specific module. c. Condensation accumulation must be considered for all enclosures. The Supplier is responsible for determining if condensate drains must be installed in cabinets for drainage (or another condensation removal method) for the specific location where the enclosure will be located.
6.21. Polycarbonate	Polycarbonate could be used with prior approval from Nestlé.
6.22. Plasticized polymeric materials	Plasticized polymeric materials (mainly PVC, used for moulds, pipes, gaskets, etc…) must be identified. Phthalates plasticizers must be replaced by alternative plasticizer(s).

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6. DESIGN – General Hygienic Rules

6.23. Area above unpackaged Product (for Medium and High Hygiene levels)

Areas above or adjacent to the unpackaged product:

- Must not be painted.
- Must not contain fasteners.
- Minimize small elements that can become loose with time; if required, then they should be fixed permanently.
- Avoid piping, wiring, and areas of waste accumulation. If unavoidable, removable stainless steel catch pans should be used to eliminate the possibility of waste or dirt falling onto the product.
- No drains are allowed above product

6.24. Cleaning Procedure

In the Machine Specific module, steel quality needs to be properly defined as it may be different depending on the cleaning procedure and the cleaning agent used (refer to the chemical supplier).

6.25. Steel Quality and Finish

All steel parts and weldments in contact with product must be:

a. Made with stainless steel 304 or 316 (to be reviewed with Nestlé) with an appropriate finish (maximum surface finish roughness Ra of 0.8 microns), unless specified differently in the Machine Specific module.
b. For Basic Requirements: typically surface finish roughness Ra of 3.2 microns.
c. All grinding and polishing of SS must be done with dedicated tools to avoid cross contamination.
d. Exterior carbon steel surfaces of all machine including controls, valves and other appurtenances shall be painted
e. Stainless steel, machined and bearing surfaces shall not be painted. Electro-polish all stainless steel product contact surfaces prior to final assembly where required.

For Medium Hygiene level

f. Typically surface finish roughness Ra of 0.8 microns.

For High Hygiene level

g. Surface finish roughness Ra should not exceed 0.8 microns.
h. In machine including CIP, steel quality should be SS 316 (as it provides a better resistance to chloride containing solutions).

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6. DESIGN – General Hygienic Rules

6.26. Steel Quality

Unless specified differently in the Machine Specific module, mild steel parts that have no direct contact with the product, should be powder coated, provided they are compatible with the product characteristics and the cleaning procedure used. All coatings should be reviewed with Nestle. Painted steel parts should be identified and approved by Nestlé

For High Hygiene level

Painted mild steel cannot be used on machine with high hygiene requirements. Use of materials other than stainless steel should be reviewed and approved by Nestle.

6.27. Vacuum & Ventilation Systems

In designing such systems (including cooling / ventilation of electrical cabinets), three basic concepts must be adhered to:

a. Air inlets shall be located on the side of the panel at the highest practical position and shall be fitted with filter units that achieve a filtration level of 3 microns maximum.
b. Components must be completely mountable so that each section can be fully cleaned.
c. Flexible connections with corrugations must be avoided.
d. Dry product filling sometimes involves use of vacuum (or air) for other purposes - if directly in contact with the product - this air must have a source designed to free it of potential contaminants (eg. lubricant and dirt), the solution must be approved by Nestlé project manager.
e. Exhaust systems must have welded seams with adequate access for cleaning and inspection.
f. Exhaust duct sections must have a drain (e.g., to the floor) to prevent drainage from going back into the machine.

6.28. Compatible Materials

a. Materials not permitted for use include wood, glass, enamelware, or belts with exposed cloth backs or centers.
b. Seals and O-rings should be chosen to be compatible with the products, cleaners, and sanitizers used on the line. . When possible, metal detectable O-rings, gaskets, and seals must be used.
c. Materials used in machine construction are compatible with the product, the environmental conditions they will be exposed to, and the cleaning methods and chemicals used to clean the machine. See machine specific module for specific chemicals used.
d. Metals used are compatible with one another. Selection of materials must consider all types of corrosion, including galvanic corrosion.

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7. DESIGN - Machine Frame

7.1. General

When designing the machine:

- Open frames are preferred as they give better visibility, cleanability and sanitation.
- No extra holes should be drilled as they will make cleaning impossible.

7.2. Frame

a. Preference must be given to solid sections to avoid creation of hollow bodies.

b. Avoid cross bracing wherever possible by designing heavier structural members with solid gussets.
Round framework is preferred for horizontal framework pieces. Where flat surfaces are used, they must be turned 45 degrees to horizontal.

For Basic and Medium Hygiene level

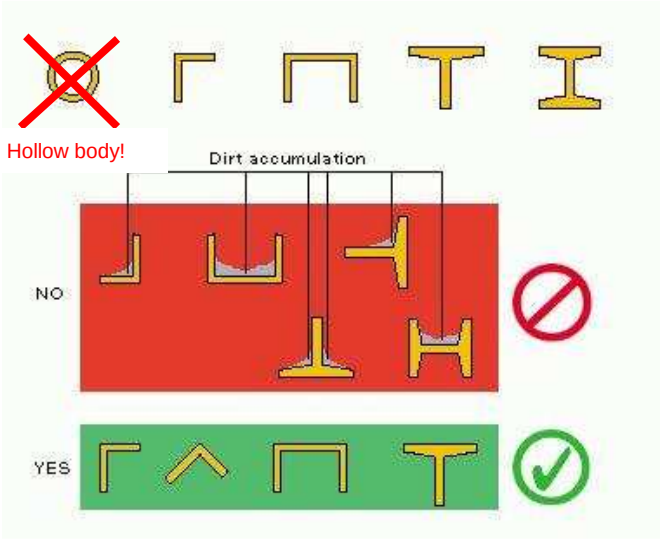
c. Unless specified differently in the Machine Specific module:
- It is OK to use painted mild-steel for the frame.

For High Hygiene level

d. Unless specified differently in the Machine Specific module:
- Aluminum or stainless steel frames should be used in dry cleaning areas.
- Use only stainless steel frames in wet cleaning areas.

7.3. Shaped Channels

They should be installed upside-down to avoid dust collection.
See example below:



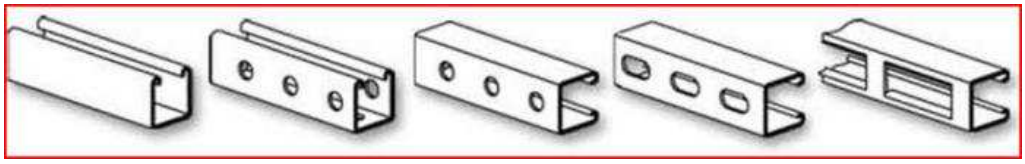
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7. DESIGN - Machine Frame

7.4. Profiles

Not to be used as impossible to clean.
(pictures are for reference only)





8. DESIGN – Guards

8.1. General

All guards must be designed to allow the operator to easily observe, inspect, and monitor the machine's operation and condition, while maintaining a safe working environment and not requiring the removal of the guards. Machinery must have "transparent" safety guards to allow inspection in working conditions. All machinery guards slope away from product zones and are easily cleaned (compliant with all safety requirements).
The material used for guarding should:

- Resist the cleaning procedure in place in the factory, including the use of chemicals where required.
- Be scratch resistant when dry cleaned (wiping), as the product dust can be very abrasive.
- Guarantee the visibility throughout the life span of the machine.
- Guards should not deflect on contact or be able to be forced or bent to allow access to machinery.

8.2. Accessibility

Guarding should be designed to provide easy and safe access for cleaning and inspection.

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8. DESIGN - Guards

8.3. Plastic Guards

Polycarbonate sheets are acceptable to be used and need to be:

- scratch resistant
- UV resistant.

They should have a minimum thickness of:

- 10 mm (3/8") for fix guarding plates
- 12 mm (0.47") for "movable" guarding doors (mounted either as fixed sheets or upon hinges).

Acrylic sheets (PMMA) are not acceptable as they may shatter.

8.4. Mesh Guards

a. Mesh guards should not be used in product filling machine that are dry cleaned.
b. Where mesh filled panels are used, the framework shall be constructed of angle section and be free from crevices and capillary gaps between assembled components.
c. For medium and high hygiene areas, laser cut stainless panels are preferred over woven wire.
d. In wet cleaning areas, mesh guards are a preferred option over plastic guards, as they are not sensitive to the cleaning agent used.

8.5. Glass Guards

Under Nestlé validation, laminated glass guards can be used in wet cleaning areas as they will have the right resistance to the chemicals used during the cleaning process and will also provide the required hygienic level (e.g. where we need over-pressure in the filling area).

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9. DESIGN - Components

9.1. General

Machine or component in excess of 50 kg (150 pounds) which is required to be periodically removed shall be equipped with lifting eyes and identified to Nestle at the quotation stage.

9.2. Conveyors

a. If belting is used as product contact surfaces, it should be non-absorbent, fully encapsulated, cleanable, and should be designed to be compatible with the cleaning methods employed at the location (wet or dry).
b. All belting is easily removable or the belt tension can be sufficiently decreased without tools so that the surfaces underneath belting can be easily accessed for cleaning.
c. Wherever possible, solid flat belt for conveyor applications will be used. Guides and stops will be provided to avoid run off causing belt abrasion against machine elements. All guides and rollers are to be of the type that will not abrade the belt.

9.3. Fasteners: Nuts, Bolts, Screws, …

a. Design of the machine should minimize the need of fasteners and prefer forming technologies that eliminate the use of fasteners.
b. If required, fasteners should not have exposed threads and will have a positive locking method to prevent falling or vibrating off., such as Nylock fasteners. Spring steel, Tinnerman type, or push-up nuts are not acceptable.
c. Nuts need to be placed on “top” of the bolts. This allows for visual inspection in the case that the nut falls off.
d. Fasteners are not allowed in machine parts having direct food contact. Where practical, all fasteners should be captured, to prevent introduction into the product stream.
e. All fasteners should be made of stainless steel.
f. Avoid the use of difficult to clean fasteners (like hexalobular, Phillips screws).
g. Where possible, hexagonal fasteners should be used as they are easier to clean.
h. Rivets should not be used for joining surfaces.
i. Setscrews, keys, wing nuts, and/or other projections, also inside threads, tapped holes, keyways, sockets, dead end pockets, or other recesses should be minimized.
j. Nuts shall not be placed so as to form pockets.
k. Where possible threads should not be exposed. If not possible, the length of exposed threads is to be no more than 3-5 threads, except for adjustment screws.

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9. DESIGN - Components

9.4. Bearings, motors, gearboxes, chain

a. Bearings, motors, chains and gearboxes should be mounted outside the open product area.
b. Motors should be easily accessible and cleanable.
c. If lubrication is required, they should use only H1 food grade lubricants (see next chapter).
d. Where possible, bearings should be of "sealed for life with food grade lubricant" type. If sealed for life, grease fittings should be not be present.
e. Pillow block and flange bearings shall have cast metal housings. Pressed or stamped metal housings are not acceptable for any type of bearing.
f. Components which contain oil must have a drain plug at the lowest point and a sufficient retention pan. Oil filling opening must be equipped with a coarse filter to prevent parts to fall into the tank during filling operation.
g. Open gears or gears covered with mesh shall not be used as they cannot be cleaned
h. Motor fans do not blow air onto or in the direction of product contact surfaces.
i. Mechanical variable speed drives are not to be used.
j. All gear boxes shall have common gear ratios wherever possible, ie; special or custom gear ratios should be avoided.
k. Cogged timing belts are to be used instead of roller chains unless approved by Nestle. The drive system shall be designed to allow for belt replacement without removing any components except machine guarding.

9.5. Hoses

a. All hoses and their support assemblies should be easily accessible and designed to be easily removed for maintenance, cleaning, and changeovers. Supports should be designed so that hoses will not abrade or be damaged in any way during the operation of the machine. Hoses should be designed to be self-draining, with no pooling of liquids. All hoses should be transparent, when possible.
b. All product hoses and their support assemblies are easily removable for cleaning. All hoses should be transparent (or allow visual inspection of the product within the hose) and the interior surfaces must meet product contact surface guidelines.

9.6. Shaft

Cantilevered shafts should be avoided where possible and if necessary, should be kept to a minimum.

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9. DESIGN - Components

9.7. Electrical Panels

a. See module 6 Electrical and Automation.

9.8. Utility & supply lines

Utility supply lines and piping are separated by a minimum 10 mm (3/8") to prevent catch points and to allow for cleaning.

9.9. Labelling

a. Labeling for pneumatic equipment must be made on the tubes and on the devices and cross-referenced with the drawings.
b. All operator buttons are to be labeled in the local market language.
c. All components shall be clearly labeled with a reference number which can be cross–referenced with the drawings.
d. All labels shall be permanently fixed and not placed on detachable part.
e. All labels must be designed for industrial use and made in type characters.

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9. DESIGN - Components

9.10. Pneumatics

a. Supplier shall supply a filter and regulator to the machine. Wherever possible pneumatic components shall be designed to operate without lubrication eliminating the need for a line lubricator. Supplier will include a lockable 'dump' valve. Valve must automatically dump air on an Emergency Stop, unless air is required to keep machine in a safe condition.
b. Main compressed air distribution piping will be schedule 10 stainless steel.
c. Where possible, compressed air piping will run along the system parallel to the cable tray.
d. All compressed air supplied to the machine is required to be filtered at the point of use. The compressed air will be filtered for particles greater than 0.1 microns if it is to be used for food contact or food contact surfaces. The ISO Quality Class of filter for systems that come in contact with product or product surface is 1.4.1. For systems without product contact a filter rating of 3.4.2 is acceptable. Vendor to confirm level of filtration, and obtain approval from Nestle Project Manager.
e. Plant air conditions are detailed in the Machine Specific module. Any additional filtration to meet the water and oil standards must also be provided at the point of use.
f. Manifold mounted valves should be used where possible. All pneumatic enclosures must contain 20% spare capacity for future expansion.
g. In situations where there are eight (8) or more valves mounted on the same manifold, compressed air will be fed from both ends of the manifold to prevent valve starvation.
h. All manifolds will contain a pressure gauge to show active air pressure in the manifold. And include red/green indicators showing proper pressure settings.
i. A sensor is required to detect low pressure conditions in the machine's main supply line. Low pressure detection will create a machine fault condition.
j. Pneumatic components will not be mounted in MCC or switchgear enclosures.

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10. OPERABILITY - Fast Changeover

10.1. Fast Changeover

A checklist and SOP must be provided by the supplier with the exact list of changes that have to be completed for a changeover to a different product or format. Times to carry out size change over on the machine must be clearly shown. The parts that shall be changed for a changeover to a different product/format shall be identified with visible color coding and numbering / referencing to facilitate their installation and storage. Same colors and coding shall be used for all the machine of the line. Nestle to provide the color scheme for change parts.

Size changing should preferably require no tools with running adjustment achieved by “electronic control”, hand wheel or some other acceptable non tool operation (fixed position datum’s). All adjustable items will be provided with calibrated scale measurements relating wherever possible to the product size being packed, for error-free change of format. Variations due to size change must also be visually managed and clearly marked.

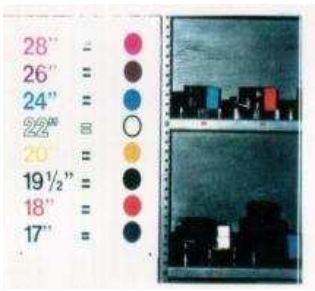
The size change-over of a machine should be designed to require less than 15 minutes for one (1) trained operator (unless otherwise specified or agreed to with the supplier), preferably using no tools. If tools are needed they should be delivered with the machine. Machine should have a provision for storage of the tools needed within close proximity of their use.

The weight of each individual change part requiring lifting by hand shall not exceed the value specified in Safety module (M4). Parts which weigh more than specified must be stamped with the weight of the part, and provide an appropriate means to support the part during change and transport.

For machines which have multiple sets of change parts, the supplier must provide a mobile storage solution for efficient changeovers and protection of the parts.



Trolley with format change parts for a filling machine with dedication location for each part.



All format parts and machine settings have their own color designation

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11. OPERABILITY - Centerlining

11.1. Centerlining

Machine must be easy for the operator to reset to standard conditions and change over parameters. Clear marking of the required standard conditions is essential (see examples below). Simple techniques should be used to indicate deviations from standards. The supplier must provide the centerlining for the FAT, which will be validated during SAT. Upon successful completion of the SAT the supplier will deliver, in writing, a list of all the centerlining settings.

All points which have to be adjusted have to be listed in the list of centerlining points with the following information:

- Where is the point on the machine located
- Description of the Centerlining point
- Picture of the centerlining point
- Function of the centerlining point and description what is happening in case the point is not adjusted correctly
- Define what effect each centerline has on losses (minor stoppages, sources of contamination, breakdowns, change over times, quality, safety)
- Frequency (per shift, daily, weekly, monthly, quarterly, yearly)
- Who has to do the adjustment (technician, operator)
- Reasons for the adjustment (changeover, abrasion, regular adjustment)
- Parameter, target value
- Unit (mm, cm, bar, ℃,…)
- Tolerance (+/-)
- Description of the centerlining or adjustment activity
- For each format one color has to be defined Nestle to provide the color scheme for each product.
- Size of matchmarkings has to be standardized and agreed upon by Nestle.



Number setting


Rulers and Matchmarking


Matchmarking

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12. MAINTAINABILITY

12.1. Autonomous Maintenance

With the Autonomous Maintenance approach, activities like Cleaning, Inspection and Lubrication (CIL) are transferred from technicians to operators. This increases ownership in machine and results in reduced breakdowns, minor stoppages, abnormalities and losses. It adds specific design considerations to make CIL easier and safer for the operator.

12.2. Cleaning, Inspection and Lubrication (CIL)

The machine must be delivered with a **CIL-checklist** that contains 3 types of tasks:

- **Cleaning:** All the places that need to be regularly cleaned with their corresponding procedures and tools.
- **Inspection:** All inspection points to check for proper machine operation.
- **Lubrication:** All lubrication points to be oiled or greased on a regular basis. The design should facilitate this task by operators as much as possible and also include a lubrication map.

The checklist for Cleaning, Inspection and Lubrication must be approved by Nestlé before delivery.

12.3. Planned Maintenance

Planned maintenance enhances predictive tasks that will lead to Zero Breakdowns and decrease overall maintenance costs in the long term.

12.4. Labeling Components

All components that require routine replacement (i.e. belts, filters, etc…) must be clearly labeled, on the machine near point of use (as per the Nestlé factory standards and using welded stainless steel tags/labels, laser or CNC engraved, subject to hygienic requirements).

12.5. Maintenance plan

12.6. Automatic Monitoring System

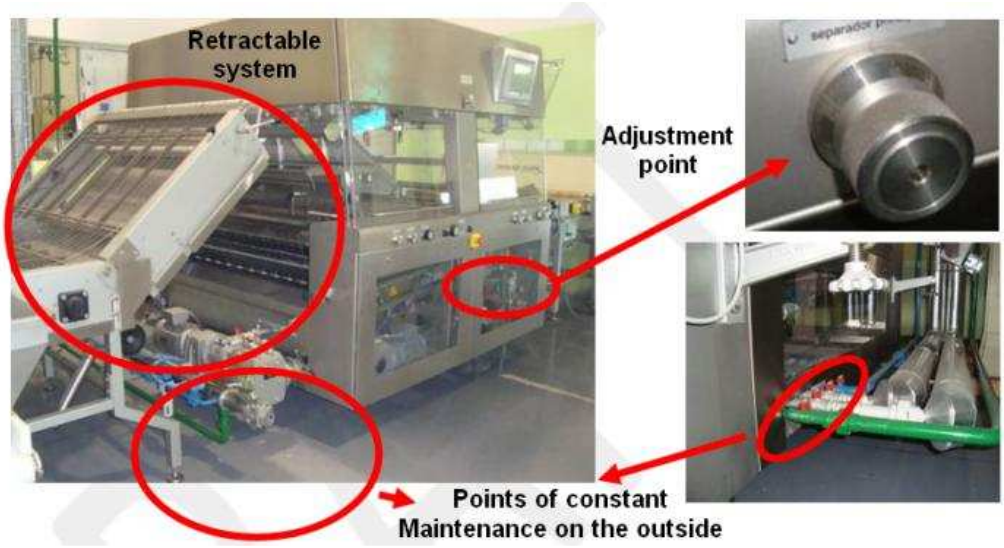
It is required or expected to have an automatic monitoring system for critical operating parameters (i.e. temperature, oil level, pressure, etc...), that triggers alarms when approaching and/or exceeding limits.

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12. MAINTANABILITY

12.7. Easy Access for Maintenance, Cleaning, Adjustments

Machine, parts or devices which require routine maintenance (more than 3 times per year), cleaning, or adjustments shall be easily accessible. Positioned on the outside or have resources facilitating their removal and replacement (e.g.: pulleys, rails, transport cars, retractable / removable systems, etc.), as in the examples below. All machine components/parts that may require routine replacement must not require removal of ancillary parts, other than guarding to exchange. This access must always be **safe**; this means that the operator is not exposed to any risk, such as climbing a ladder.
Access to wear parts must facilitate replacement within one hour or less (includes bearings, belts, chains, clutches, etc.).



12.8. Vibration monitoring

a. Supplier will supply the ability to check vibration signatures. Supplier will identify and mark transducer location for predictive maintenance purposes.
b. Supplier will provide documented baseline vibration signatures at the end of the commissioning phase.

12.9. Thermal

Supplier will provide access to each motor for temperature monitoring during steady-state operation, without compromising safety guidelines. Supplier will identify and mark location for predictive maintenance purposes.
Supplier will provide documented temperature readings at the end of the commissioning phase.

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13. MAINTAINABILITY – Lubricants

13.1. General

a. When designing the machine:
   - Prefer “grease free” solutions.
   - Greasing locations should be clearly marked and labeled to avoid any confusion between the different types of lubricants.
   - No dripping of lubricant is allowed.

b. Machine should be designed such that lubricants do not come into direct or indirect contact with products.

c. All lubricants have to be food grade H1 level.

13.2. Food grade lubricants

a. Lubricants are defined as “Food grade” when they:
   - Do not cause any physiological hazard (provides safety)
   - Do not affect the product odor and taste in any way
   - Do not affect the quality of the product.

b. In any case, food grade lubricants need to be registered with NSF (National Sanitation Foundation, USA) as H1 and certification provided to Nestlé.

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14. CLEANABILITY

14.1. General

The machine design shall:

- Allow for safe and easy cleaning of all areas in/on the machine (easy inspection and easy accessibility) within the time specified under Machine Specific Module.
- Avoid “hard to reach” places.
- Provide access to areas underneath and around machine for cleaning and maintenance.
- Describe all the cleaning procedures in detail (how to clean, using which tool and cleaning agent, at which temperature, at which frequency …).
- Where applicable, describe the decontamination procedure.
- Where access or disassembly is not possible, the entire assembled unit is cleanable using techniques that eliminate product risks (like CIP).
- Identify places that are located inside the machine and that can become dirty because of product residue or grease. Ensure proper lighting of all parts that need inspection and cleaning.
- Be designed and installed with consideration to its particular type of cleaning and maintenance requirements.
- Ensure that all materials of construction are compatible with required cleaning, this may include: use of caustic/acid, sanitizing, or line lubrication solutions.
- All parts will remain intact without changes in shape, structure & function through cleaning & sanitation protocols.
- The cleanability of the machine, including CIP where appropriate, must be demonstrated per Nestlé’s cleaning procedure and validation.
- Construction of machine must meet the GMP definition of ‘adequately cleanable’ [21CFR110.40].
- Nestlé needs to inform the supplier if any special cleaning process is used (based on product requirements or local regulations)

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14. CLEANABILITY continued

14.2. Wet Cleaning, CIP, External Foaming (for High Hygiene level)

a. Additionally, in areas where wash-down is required, the machine should incorporate following hygienic design features:
   - Design, and installation, should prevent any harborage points for bacteria. This includes all parts of the machine. Generally 10mm (3/8") stand-off between adjacent flat surfaces. This includes motors and other field devices such as photoeyes, solenoid valves, switches, brakes, etc.
   - Be completely self-drainable. The required minimum draining gradient is 1cm/m (1% slope).
   - All materials used shall be Stainless Steel**.**
   - Other materials such as brass, aluminum, must be approved by Nestlé.
   - Frames/legs must be of stainless steel and polished to a surface finish of 3.2 microns.
   - Machine installed in a wash-down area shall be capable of being washed down with high-pressure hoses including all electrical panels and sensors. Machine should be rated at NEMA 4X or IP 66 minimum. The machine must be capable of withstanding the frequency and the cleaning procedure specified in Machine Specific module.
   - All mechanical and pneumatic panels located within the wash-down area, will be stainless steel construction.
   - Pneumatic tubes shall enter the enclosure through bulkhead fittings from underneath.
   - All bearings in wash-down situations shall be stainless steel with stainless steel housings.
   - Guarding that requires visibility are to be made of stainless steel perforated sheet or wire mesh. Openings in guarding are dictated by the in module 4, minimum safe distance - opening.
   - If transparent plastic sheet to be used, then they should not get opaque due to cleaning procedure or chemicals used.
   - Guarding support frames must be made from solid Stainless Steel and designed in a manner to support the guarding and mounting hardware without deflection.
   - If belting is used, it should be non-absorbent, fully encapsulated, cleanable and should be compatible with the cleaning agents specified in Machine Specific module.
   - All pipes must be installed:
     - With a slope of 3% towards the drainage point
     - Without water or air-pockets
     - Without dead ends longer than 3x pipe diameter.

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14. CLEANABILITY

Wet Cleaning, CIP, External Foaming (for High Hygiene level))	To ensure a proper CIP process, it is important to clarify roles and responsibilities between Nestlé and the supplier, including the signals exchanged between the Nestlé factory CIP process and the machine.

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15. CIL - Cleaning

15.1. Cleaning

The machine must be easy to clean by the operator:

- The machine design allows easy access and easy cleaning without water in dry areas. In wet areas the machine has proper drain devices and anti-corrosive materials.
- The design prevents dust accumulation corners or hollow bodies.
- The machine is provided with a cleaning instruction.
- Disassembly of machine for cleaning shall be designed for no (or minimum) tools required. If tools are required (not preferred), then approval must be given by Nestle and the fastener and tool size should be common across all parts

15.2. Avoid Hard to reach places (HTRP)

Easy to clean means that the machine does not contain Hard to Reach Places (HTRP) as shown in the picture.



15.3. Avoid Sources of Contamination (SOC)

The machine must be designed to prevent the arise or grow of possible Sources of Contamination (SOC). An example of SOC is excessive lubrication:



Excessive lubrication



Bad example of installation of a motor



Aseptic motor drives (good example)

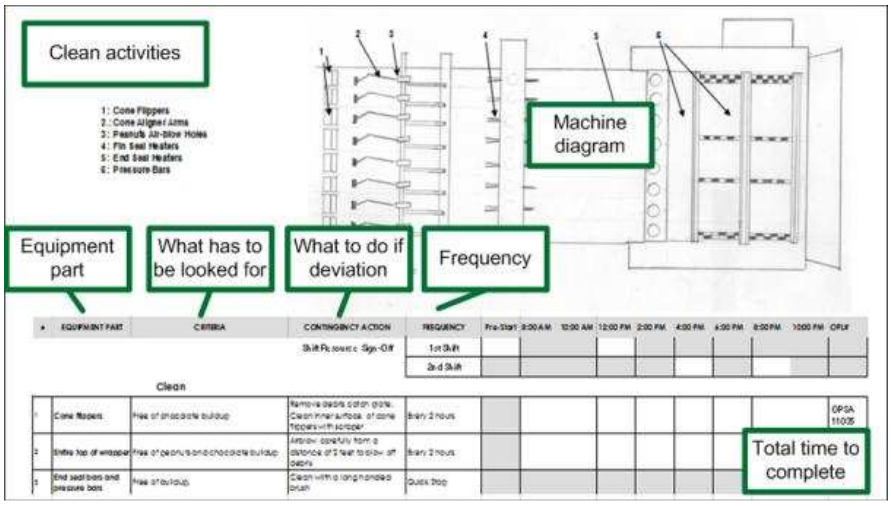
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15. CIL – Cleaning continued

15.4. Cleaning instruction

The CIL-checklist must contain a detailed cleaning instruction, with required tools, cleaning detergents, method of cleaning, frequency and duration.
The supplier to provide the following:

- Picture of the point / area, which has to be cleaned
- Description of the cleaning activity
- Detailed description of the used cleaning method. Preferable the cleaning is done dry.
- Information regarding Log out Tag out and Personal protection equipment (PPE)
- Tools which have to be used (and agreed) for the cleaning activity
- Cleaning detergent (preferable from the list of preferred Nestlé detergents). The data sheet for the detergent has to be provided and checked and approved prior to the use by the quality manager and the SHE manager
- Frequency (per shift, daily, weekly, monthly, quarterly, yearly)
- Number of people needed
- Cleaning time:
    - Detailed time needed for each single cleaning activity
    - Maximum time for cleaning activity (specified in Machine Specific module)
- Education and training needed for the cleaning activity.
- Information if cleaning activity is carried out during runtime / downtime



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16. CIL - Inspection

16.1. Inspection

The critical parts of the machine that need to be inspected must be visible (transparency/light) and easy to reach. Preferably no tools are needed to perform the inspection. These include but are not limited to:

- Belts
- Chains
- Pulleys
- Gearboxes
- Oil reservoirs
- Filters

Below is an example of a transparent safety guard that allows operators to see how the components are working and) anticipate corrective actions before breakdown:





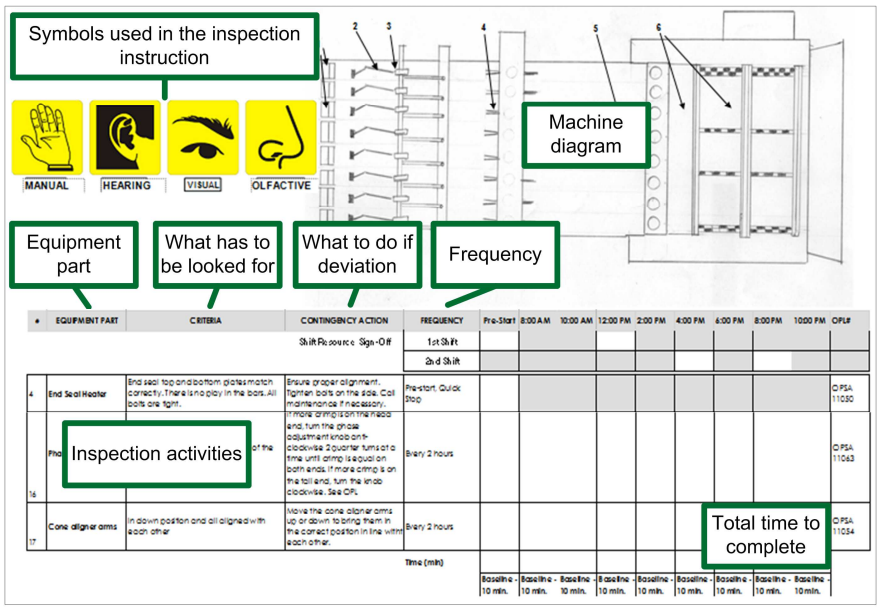
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16. CIL – Inspection

16.2. Inspection checklist

The CIL-shall include a checklist with all important inspection points specifying standards, abnormalities and inspection frequency. The checklist includes at least:

- Picture of the point / area / machine part, which has to be inspected
- Description of the inspection activity
- Information regarding Lock out Tag out and Personal protection equipment (PPE)
- Tools which have to be used for the inspection activity and how the inspection is performed, ie; visual, touch, sound or smell
- Frequency (Per shift, daily, weekly, monthly, quarterly, yearly)
- Number of people needed
- Time needed for the inspection activity
- Education needed for the inspection activity
- Information if inspection activity is carried out during runtime / downtime
- The inspection point needs to be marked (see examples below). For each inspection point according to Nestle labeling standards in this document.



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16.3. Lighting for inspection

Critical components that are shaded by the machine frame, sub-assemblies, and/or guarding must be lit using lighting fixtures. Lights must be switched on and off from outside the guarding. Lighting must be specified and approved by Nestlé (see Module 4 - SHE). LED lights are preferred. Vendor must provide options for enabling the lights during LOTO. To be discussed with the Nestle project engineer.

16. CIL - Inspection continued

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16.4. Visual controls for inspection and ease of center-lining

The design must allow easy checking of working conditions, visible from the operator's position. The machine shall be provided with digital displays, dial indicators, colour coding, scales with pointers, etc., suitable to provide easy visual inspection and indication of the reference settings and relevant acceptable range. The design allows inspection of machinery through transparent guards, without having to open cabinets or unbolt steel protections. All the instruments, drives, control buttons and points of adjustment or checking must be positioned on the outside of the machine, at a height of easy access and/or viewing for the operator.
The operating condition is colour indicated:

- Green = Good condition
- Amber = Warning, plan action (not for centerlining)
- Red = No good condition, immediate action required

Following examples must be applied:

All filters must have an indicator to show when the filter element requires replacement. If possible the machine should be designed for filter replacement while the machine is running.



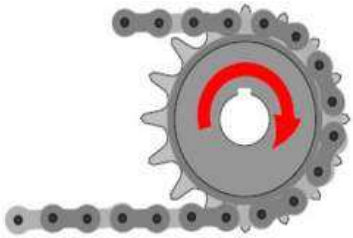
Indicate right position of belt tensioning device



Easy adjustable visual control of work range at measuring instruments



Indicate a belt deviation with red marking



Indication rotation direction

16. CIL - Inspection continued

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16.5. Visual controls for inspection and ease of center-lining

Following examples must be applied:



Flow direction of material must be indicated.



Piping labeling must be according to DIN 2403

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17. CIL - Lubrication

17.1. Lubrication	The machine must be designed to allow easy and error-free lubrication performed by operators. Lubrication points must be labelled as specified in this document, visible and easy to reach. Whenever possible a centralized or automatic lubrication system must be used. At installation, the supplier shall check all motors for lubrication, and where necessary, lubricate them with Nestle approved lubricant. The supplier shall be responsible for furnishing and applying lubrication on any other machine components that require lubrication for proper operation prior to start-up. Color coding of grease fittings shall match Nestle standards and identify type and frequency of lubrication.
17.2. Over lubrication	To prevent over-lubrication, the supplier's bearing lubrication volume requirements and recommended grease gun sizes must be furnished.
17.3. Easy access for lubrication	All grease points must be piped to easily accessible clusters of lubrication nipples, color coded to denote frequency of attention required. Guards must be designed and installed such that they do not obscure grease points, oil filling or oil level inspection points. Where necessary the guarding must have a hole through which the grease nipple can be safely accessed, in conformance with Nestle safety standards and applicable codes. Oil drain points should be piped to a location where the oil can be easily collected into a container. When a centralized lubricant piping system is not installed, automatic lubricators, installed in easily accessible areas, are preferred over manual feeding.
17.4. Lubrication in running mode	Where feasible, the design permits safe lubrication while the machine is in operation.

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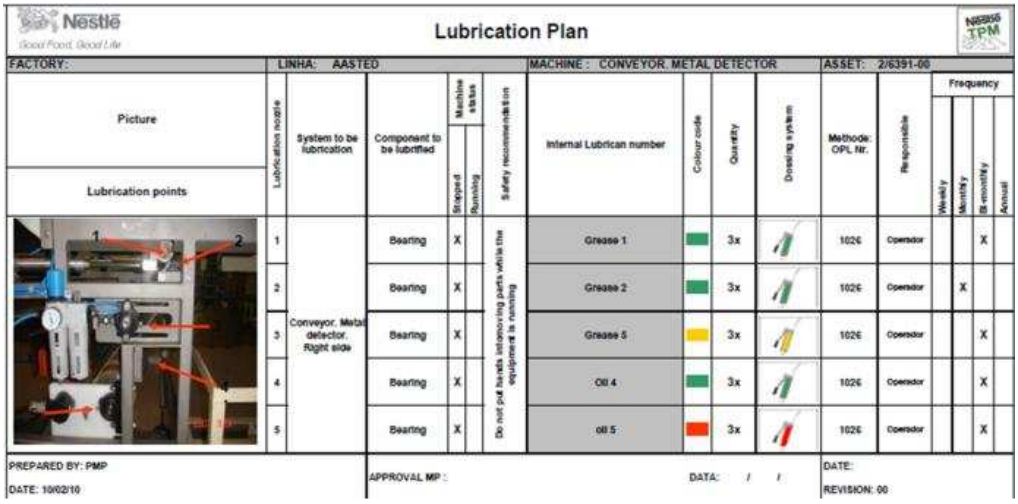
17. CIL – Lubrication

17.5. Lubrication checklist

The CIL plan-shall include a checklist with all important lubrication points specifying standards, abnormalities and inspection frequency. The checklist includes at least:

The plan includes at least:

- Picture of the lubrication point
- Description of the lubrication activity
- For each lubrication activity the exact amount of lubricant needs to be provided in order to reduce breakdowns and SOC´s
- Information regarding Lock out Tag out and Personal protection equipment (PPE)
- Tools which have to be used for the lubrication
- Frequency (Per shift, daily, weekly, monthly, quarterly, yearly)
- Information about intended use of lubrication grease. H1 grease has to be used. Preferred Lubricant: (see preferred suppliers)
- Number of people needed
- Time needed for the lubrication activity
- Education needed for the lubrication activity
- Information if lubrication activity is carried out during runtime or downtime



Lubrication Plan

FACTORY: | LINHA: AASTED | MACHINE : CONVEYOR. METAL DETECTOR | ASSET: 2/6391-00

Picture / Lubrication points	Lubrication nozzle	System to be lubrication	Component to be lubrified	Machine status: Stopped	Machine status: Running	Safety recommendation	Internal Lubrican number	Colour code	Quantity	Dosing system	Methode OPL Nr.	Responsible	Frequency: Weekly	Monthly	Bi-monthly	Annual
	1	Conveyor. Metal detector. Right side	Bearing	X		Do not put hands inbetween parts while the equipment is running	Grease 1		3x		102€	Operador			X	
	2		Bearing	X			Grease 2		3x		102€	Operador		X		
	3		Bearing	X			Grease 3		3x		102€	Operador			X	
	4		Bearing	X			Oil 4		3x		102€	Operador			X	
	5		Bearing	X			oil 5		3x		102€	Operador			X	

PREPARED BY: PMP
DATE: 10/02/10
APPROVAL MP :
DATA: / /
DATE:
REVISION: 00

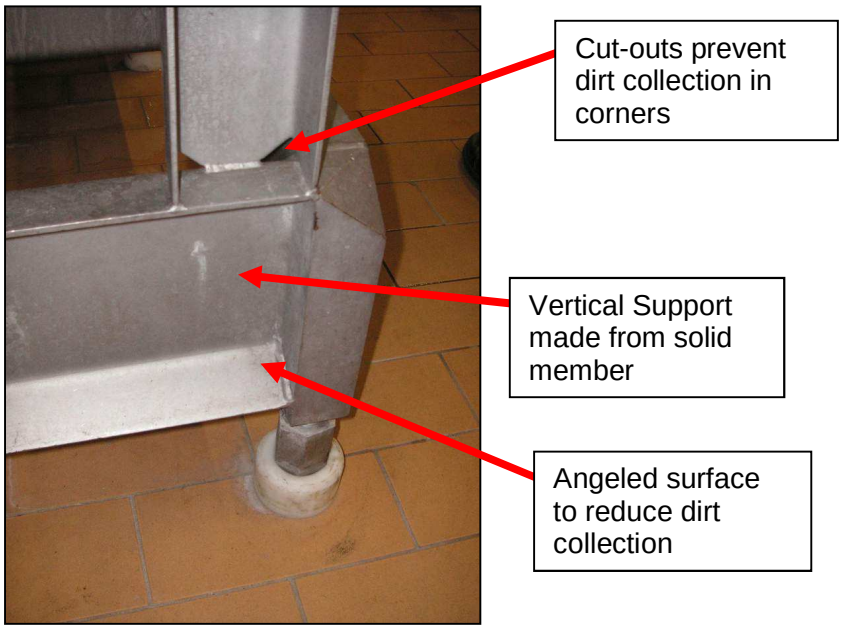
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18. Pictures

18.1. General

Below pictures are only to illustrate some requirements specified in this document.
They are not an exhaustive list of points that need to be executed.

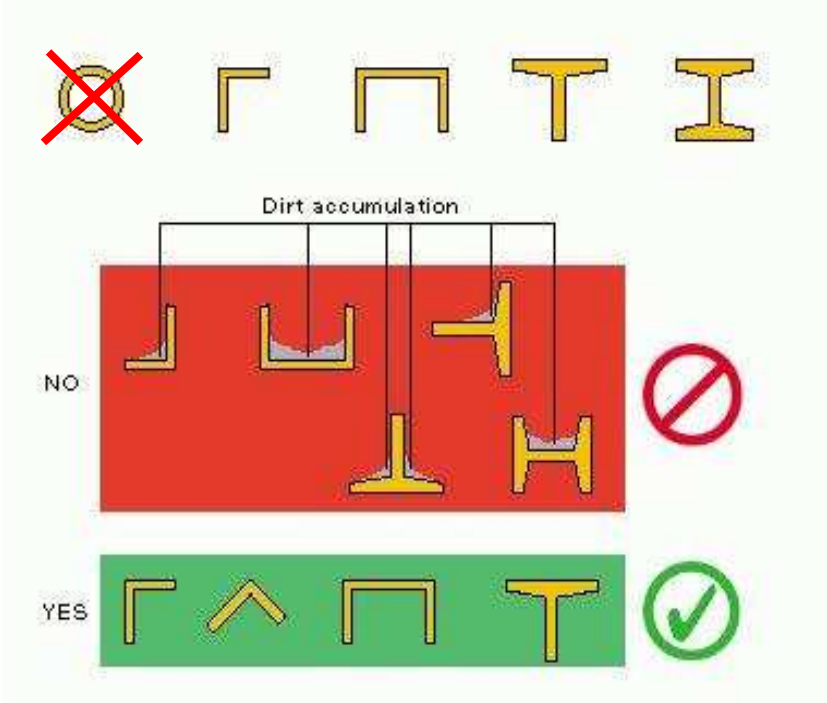
18.2. Welding & Cleanability



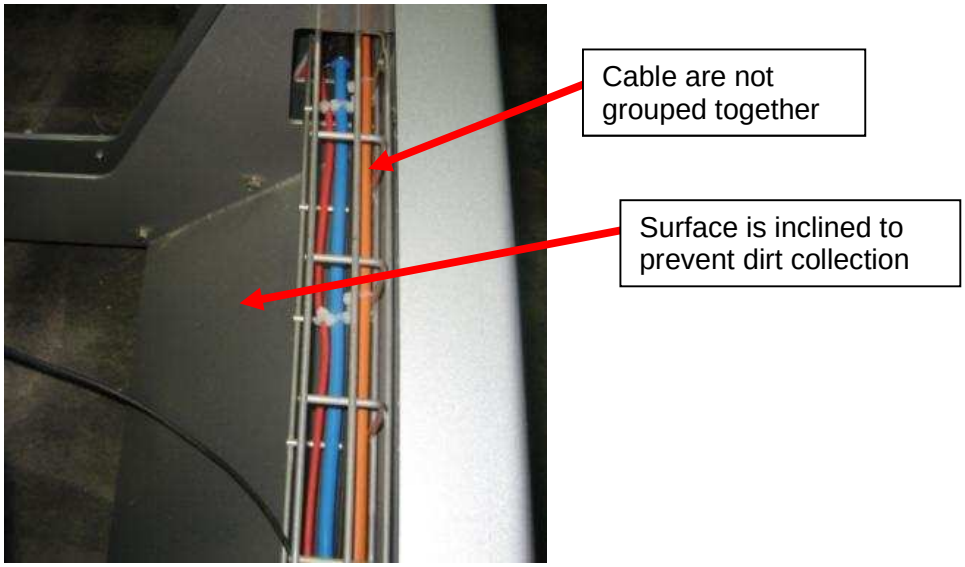
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18. Pictures

18.3. Shaped Channels



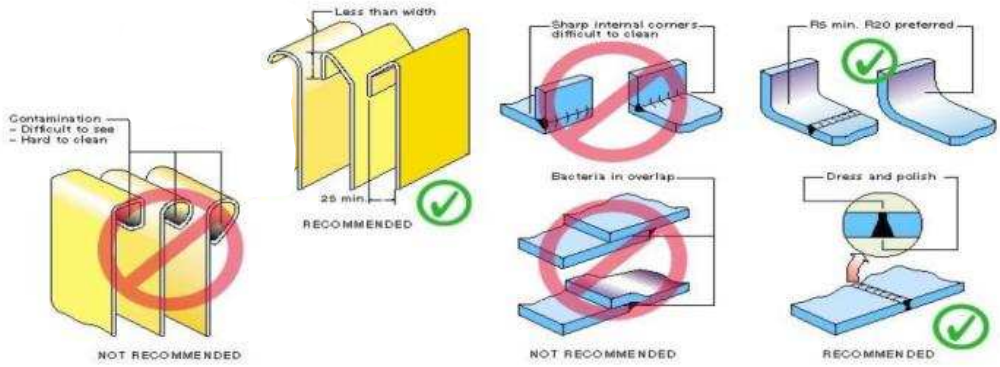
18.4. Cable Tray and Inclined surface



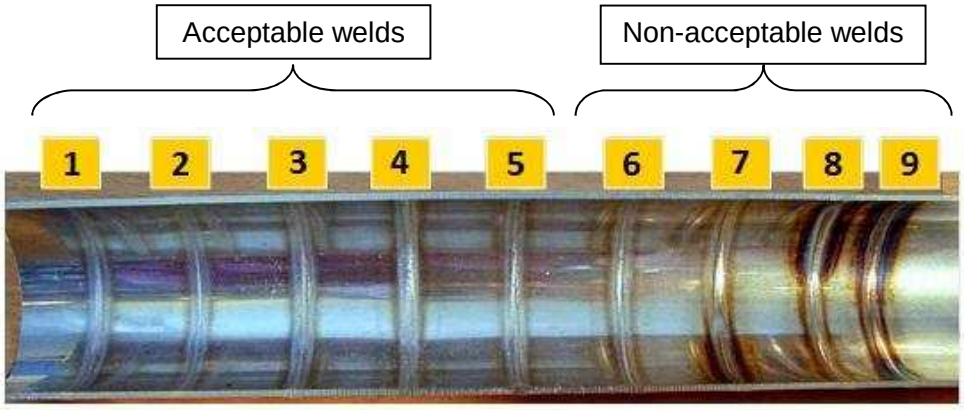
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18. Pictures, Continued

18.5. Welding and cleanability



18.6. Examples of stainless steel welding



Continued on next page

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18. Pictures, Continued, Continued

18.7. Frames examples

Fully welded structure tubes only permitted in medium and low hygiene areas. Holes or drilling to fix items is not permitted in any zone.



Continued on next page

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19. Pictures, Continued, Continued, Continued

Frames examples (continued)

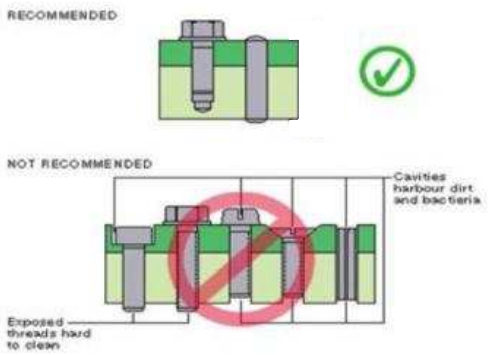
Aluminum hollow profiles are not permitted.



Hollow bodies in all areas are not permitted, e.g. conveyors feet and frames.







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20. Packing

19.1. Packing

Goods should not be dispatched without the shipping instruction from Nestlé.

a. The system components shall be properly crated or skidded, for shipment to prevent damage. Unless otherwise instructed, all crating, tarping, and securing of the machine is the responsibility of the supplier.
b. Machine shipped via air or sea freight shall be export crated and fumigated in accordance with local import regulations.
c. Pictures of the goods inside trucks or containers must be sent by email.
d. All machine must be protected during shipment, so that it arrives in a hygienic, sanitary condition. Appropriate measures must be taken to avoid corrosion during shipment. Blank parts should be wrapped into anti-rust paper or properly greased/waxed. Electrical panels and elements requiring higher protection to be sealed into appropriate aluminum sheets with air removed and desiccant incorporated into the package to absorb moisture
e. Rigging points for offloading machine must be provided. Instructions for offloading must be provided and rigging points to be identified. Bracing and supports for shipping must not penetrate hollow bodies.
f. Is it preferred that wood products used in the crating of the machine for transport shall not contribute to deforestation. Consideration is given to machine manufacturers using certified materials from sustainably managed origin (e.g. FSC). If packaging is partially or wholly wooden, use wood treated as per IPPC-standard / ISPM 15 regulation. This is not applicable to cardboard boxes.
g. Neither invoice nor packing list from the Supplier must be attached to the parcel, unless otherwise instructed.
h. Nestlé must be informed at the acceptance of the purchase order if the goods do not fit into standard containers. Machine is oversized if the gross dimensions are above the following figures: length: 12.00 m; width: 2.30 m; height: 2.50 m.

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